

Percentage Basics

A Comprehensive 2-Page Study & Reference Guide

The word **Percentage** originates from the Latin phrase "*per centum*", which literally translates to "**by the hundred**" or "**out of 100**". It is a mathematical concept used to express a number or ratio as a fraction of 100. Percentages are represented globally by the symbol **%**.

For example, if a student scores 85% on an exam, it means they earned 85 points for every 100 points possible. Percentages allow us to normalize different scales so that comparisons can be executed effortlessly.

The Core Mathematical Formula

To find the percentage of any given portion relative to a whole total value, use the foundational formula below:

$$\text{Percentage} = \left(\text{Part} / \text{Whole} \right) \times 100 \%$$

Example 1: Basic Evaluation

Question: Out of 50 students in a class, 15 are wearing spectacles. What percentage of students wear spectacles?

Solution: Here, the Part = 15, and the Whole = 50.

$$\text{Percentage} = (15 / 50) \times 100 = 0.3 \times 100 = 30\%$$

Crucial Conversions (Fractions, Decimals, & Percents)

Fluency in percentages requires knowing how to pivot dynamically between fractions, decimals, and percentage values:

- **Percent to Decimal:** Remove the **%** sign and divide by 100 (move the decimal point two places to the left).

$$75\% = 75 / 100 = 0.75$$

- **Decimal to Percent:** Multiply by 100 and annex the **%** sign (move the decimal point two places to the right).

$$0.425 \times 100 = 42.5\%$$

- **Percent to Fraction:** Place the percentage value over a denominator of 100 and reduce to simplest terms.

$$40\% = 40 / 100 = 2 / 5$$

Essential Conversions Lookup Table

Memorizing these common fractional breakdowns speeds up mental calculations dramatically in everyday contexts:

Fraction	Decimal Equivalent	Percentage Equivalent
1 / 2	0.5	50%
1 / 4	0.25	25%
3 / 4	0.75	75%
1 / 5	0.2	20%
1 / 10	0.1	10%
1 / 3	0.333...	33.33%
2 / 3	0.666...	66.67%
1 / 8	0.125	12.5%

The Three Core Types of Percentage Problems

Almost every percentage question boils down to finding one of three variables in the relationship: the *Part*, the *Percentage*, or the *Whole Total*.

Type A: Finding the Part (What is X% of Y?)

$$\text{Part} = (\text{Percentage} / 100) \times \text{Whole}$$

Example 2

Question: Calculate 20% of \$150.

Solution: $\text{Part} = (20 / 100) \times 150 = 0.20 \times 150 = \30 .

Type B: Finding the Percentage (What % of Y is X?)

$$\text{Percentage} = (\text{Part} / \text{Whole}) \times 100$$

Example 3

Question: What percentage of 80 is 20?

Solution: $\text{Percentage} = (20 / 80) \times 100 = 0.25 \times 100 = 25\%$.

Type C: Finding the Whole Total (X is Y% of what number?)

$$\text{Whole} = (\text{Part} / \text{Percentage}) \times 100$$

Example 4

Question: 60 is 30% of what total value?

Solution: $\text{Whole} = (60 / 30) \times 100 = 2 \times 100 = 200$.

Percentage Increase and Decrease

To quantify how much a value has changed over time relatively, we track percentage shifts using this explicit formula:

$$\text{Percentage Change} = \left((|\text{New Value} - \text{Original Value}| / \text{Original Value}) \times 100 \right) \%$$

- If the New Value is **greater** than the Original Value, it is a **Percentage Increase**.
- If the New Value is **less** than the Original Value, it is a **Percentage Decrease**.

Example 5: Real-World Retail Application

Question: A pair of shoes originally priced at \$80 is marked down on sale to \$60. What is the percentage markdown (decrease)?

Solution: Original Value = 80; New Value = 60. Difference = $80 - 60 = 20$.

Percentage Change = $(20 / 80) \times 100 = 0.25 \times 100 = 25\%$ decrease.

Mental Math Shortcuts

To compute percentages rapidly without scratchpad paper or calculators, utilize these scaling rules:

- **To find 10%:** Simply shift the decimal point one place to the left (e.g., 10% of 340 is 34).
- **To find 1%:** Shift the decimal point two places to the left (e.g., 1% of 340 is 3.4).
- **To find 5%:** Calculate 10% first, then divide that result by 2.
- **To find 15%:** Calculate 10% and add it to 5% of that same number.